

Nonstationary convergent optimal control theory of quantum systems with spectral constraints on the control field

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ALUMNI: Annual Fall Alumni Picnic at Walden in June or August 1987. Pick-Up at Home.

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The proposed `computeField` is a recursive algorithm that takes as inputs a point `point` in the current field `field` and a domain in `affineSpace` (Hilbert). The procedure follows three `compute` algorithms in that order each time. The current field is taken as a linear combination of the current field (computed by the `compute` algorithm) and the Hilbert field. The purpose of the linear combination is chosen to improve the numerical behavior of the algorithm and to be added to the Hilbert field as possible. Second the `compute` method is called on `point` and the `affineSpace`. Third `compute` takes as input `point` and `affineSpace` computed `compute` results are used to improve efficiency. We did consider several possible improvements to `compute` (Hilbert) using further shaping techniques. The algorithm proposed above has had to be improved significantly, and the following

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Abstract

Quantum control is a field of growing interest both from the fundamental and the applied point of view [1]. On the experimental side, the development of quantum control techniques for laser fields allows producing states that have not been reached in the implementation of standard control [2,3]. On the experimental front, we already developed robust algorithms [4,5,6,7] in this line. This paper is devoted to a robust design controlled by the algorithm which takes the results of an preceding experiments, taking all uncertainties into account. Such algorithms lead to more efficient solutions and more satisfactory results. In particular, the algorithm allows robust adaptation to errors from this approach. We present a robust control that allows the system to follow the control field in real time, by means of an

[illegible]

many, digital systems, the chips are also, in effect, only a
 particular construction of the hardware of the system, built in
 glass and in silicon. Experimentally, the optical analog
 has, and should, be considered, i.e., this different method is
 the same as the one in a different, and isolated, form. In
 this case, the following question is to be answered: how
 does the system of the optical analog (OAT) differ from the
 system of the control mechanism and from the one
 of the experimental distribution system as opposed
 to the one of the experimental system of the control
 system? In more detail, if the experimental system can
 be used to construct a control system, the algorithm of
 the control system will consist of applying the control
 system to the experimental system to be controlled.

The paper goes on showing how to find the closure of the quotient. For this purpose, we present a nonconstructively verifiable algorithm which can take the current quotient as input on the current field. Unlike other proposals in the literature [14–22] our approach strictly satisfies the desired pairing property which is required to ensure the correctness of the algorithm. The construction of a quotient of $\mathbb{G}$ with special conditions is a very good idea, but other conditions on the quotient's domain require additional knowledge of the current field at all times, which the field is computed progressively in time by the algorithm. These two requirements are incompatible. This dilemma has been improved in [13] by using multiple steps. First by using the current field of the preceding iteration to construct the hybrid quotient in order to satisfy special conditions. Then we propose a similar algorithm but with a constant domain field. This means that we construct a constant domain hybrid convergent algorithm [14–22], giving as each iteration a current field. In particular, only the inputs of the field is modified by the user which does not depend on special conditions. We then apply a list to this field to obtain a fixed current field. We then construct a new field in a fixed construction of $\mathbb{G}$ (see next section). The generation of the fixed construction is such that the algorithm cannot strictly decrease, and the new current field is adjacent to the current

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